PVsyst V8.0.18

PVsyst - Simulation report

Grid-Connected System

Project: New Project

Variant: New simulation variant

No 3D scene defined, no shadings

System power: 13.20 kWp

Udubaddawa Ihalagama - Sri Lanka

PVsyst TRIAL

Author

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VC0, Simulation date:
26/11/25 15:50
with V8.0.18

Project summary

Geographical Site
Ududaddawa Ihalagama
Sri Lanka

Situation

Latitude	7.48 °(N)
Longitude	79.97 °(E)
Altitude	49 m
Time zone	UTC+5.5

Project settings

Albedo	0.20

Weather data
Ududaddawa Ihalagama
Meteonorm 8.2, Sat=29% - Synthetic

System summary

Grid-Connected System

No 3D scene defined, no shadings

Orientation #1
Fixed plane
Tilt/Azimuth 15 / 0 °

Near Shadings
no Shadings

User's needs
Unlimited load (grid)

System information

PV Array

Nb. of modules	44 units
Pnom total	13.20 kWp

Inverters

Nb. of units	1 unit
Total power	15 kWac
Pnom ratio	0.88

Results summary

Produced Energy	20970 kWh/year	Specific production	1589 kWh/kWp/year	Perf. Ratio PR	82.64 %

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General parameters

Grid-Connected System — **No 3D scene defined, no shadings**

Orientation #1
Fixed plane
Tilt/Azimuth 15 / 0 °

Models used
Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

Horizon
Free Horizon

Near Shadings
no Shadings

User's needs
Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer	Generic
Model	TSM-DEG5-(II)-300
(Original PVsyst database)	
Unit Nom. Power	300 Wp
Number of PV modules	44 units
Nominal (STC)	13.20 kWp
Modules	2 string x 22 In series
At operating cond. (50°C)	
Pmpp	11.92 kWp
U mpp	642 V
I mpp	19 A

Inverter

Manufacturer	Generic
Model	SUN2000-15KTL-M2
(Original PVsyst database)	
Unit Nom. Power	15.0 kWac
Number of inverters	2 * MPPT 50% 1 unit
Total power	15.0 kWac
Operating voltage	160-950 V
Max. power (=>55°C)	16.5 kWac
Pnom ratio (DC:AC)	0.88
No power sharing between MPPTs	

Total PV power

Nominal (STC)	13 kWp
Total	44 modules
Module area	72.4 m²
Cell area	64.8 m²

Total inverter power

Total power	15 kWac
Number of inverters	1 unit
Pnom ratio	0.88

Array losses

Thermal Loss factor
Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses
Global array res. 581 mΩ
Loss Fraction 1.50 % at STC

Module Quality Loss
Loss Fraction -0.75 %

Module mismatch losses
Loss Fraction 2.00 % at MPP

IAM loss factor
Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.963	0.892	0.814	0.679	0.438	0.000

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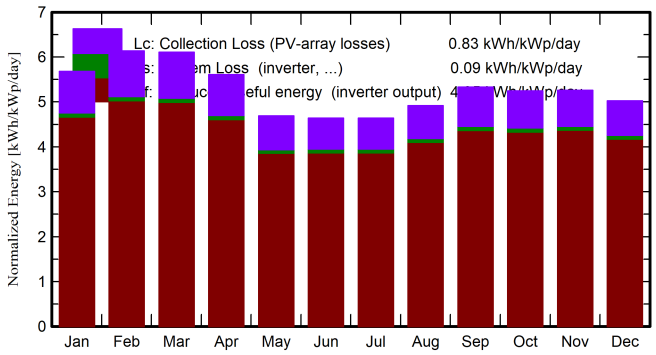
Main results

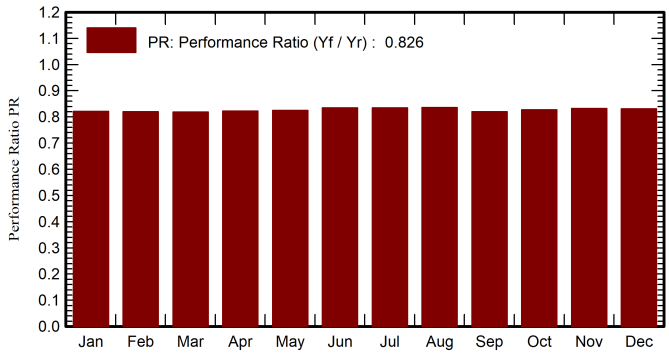
System Production

Produced Energy	20970 kWh/year	Specific production	1589 kWh/kWp/year
		Perf. Ratio PR	82.64 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor kWh/m²	DiffHor kWh/m²	T_Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray kWh	E_Grid kWh	PR ratio
January	158.4	62.61	26.35	176.2	172.6	1948	1911	0.822
February	160.3	68.61	27.11	171.8	168.5	1895	1860	0.820
March	186.4	82.76	28.12	189.2	185.4	2083	2044	0.818
April	174.4	79.05	27.85	168.3	164.4	1862	1826	0.822
May	158.2	77.23	28.84	145.3	141.1	1614	1581	0.824
June	153.4	83.51	27.98	139.2	135.0	1566	1534	0.835
July	157.1	84.92	28.20	143.9	139.7	1618	1585	0.834
August	160.5	94.66	28.00	152.4	148.4	1716	1681	0.836
September	160.8	65.54	27.38	159.9	156.5	1767	1730	0.820
October	155.2	78.76	27.25	162.5	159.3	1812	1775	0.827
November	145.3	71.10	26.21	157.7	154.8	1768	1732	0.832
December	140.9	70.80	26.36	155.8	152.9	1746	1710	0.831
Year	1910.7	919.56	27.47	1922.3	1878.7	21395	20970	0.826

Legends

GlobHor	Global horizontal irradiation	EArray	Effective energy at the output of the array
DiffHor	Horizontal diffuse irradiation	E_Grid	Energy injected into grid
T_Amb	Ambient Temperature	PR	Performance Ratio
GlobInc	Global incident in coll. plane		
GlobEff	Effective Global, corr. for IAM and shadings		

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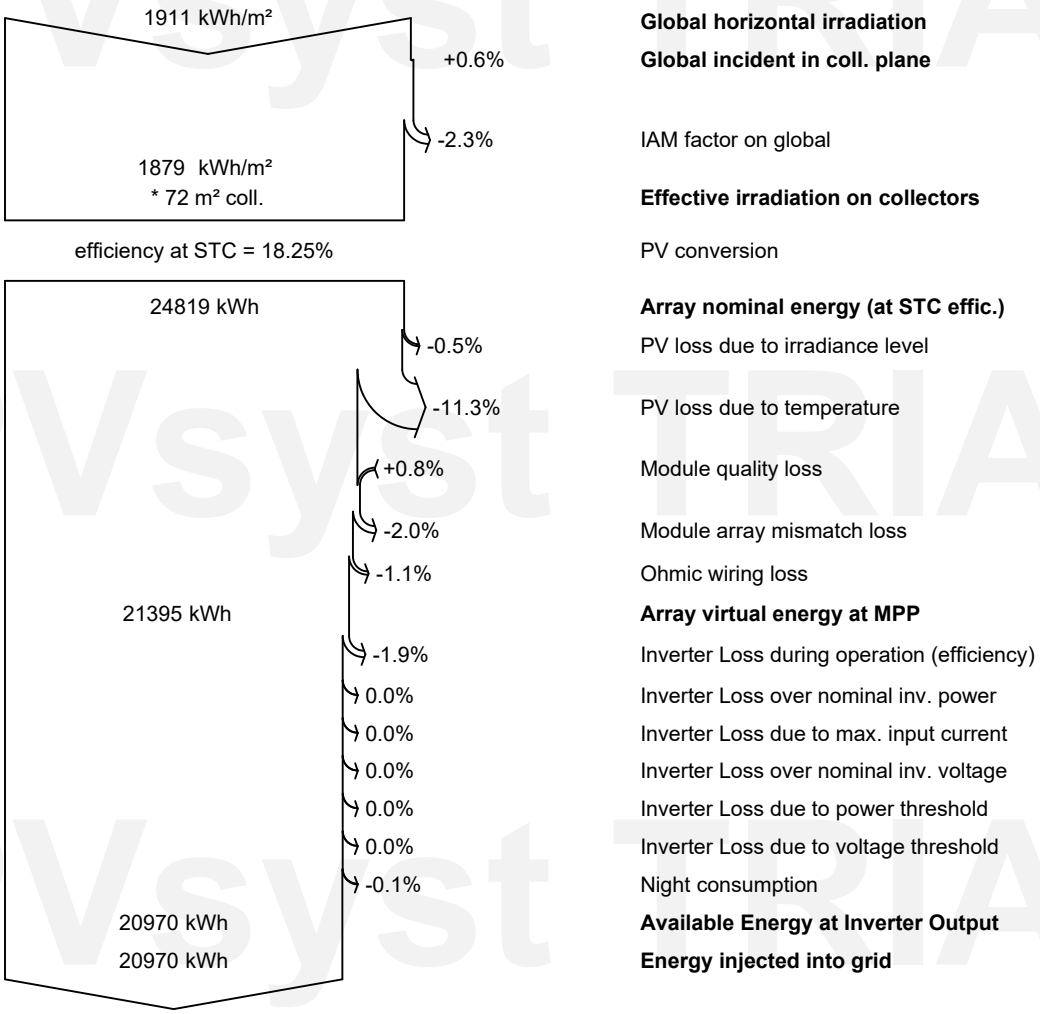
Loss diagram



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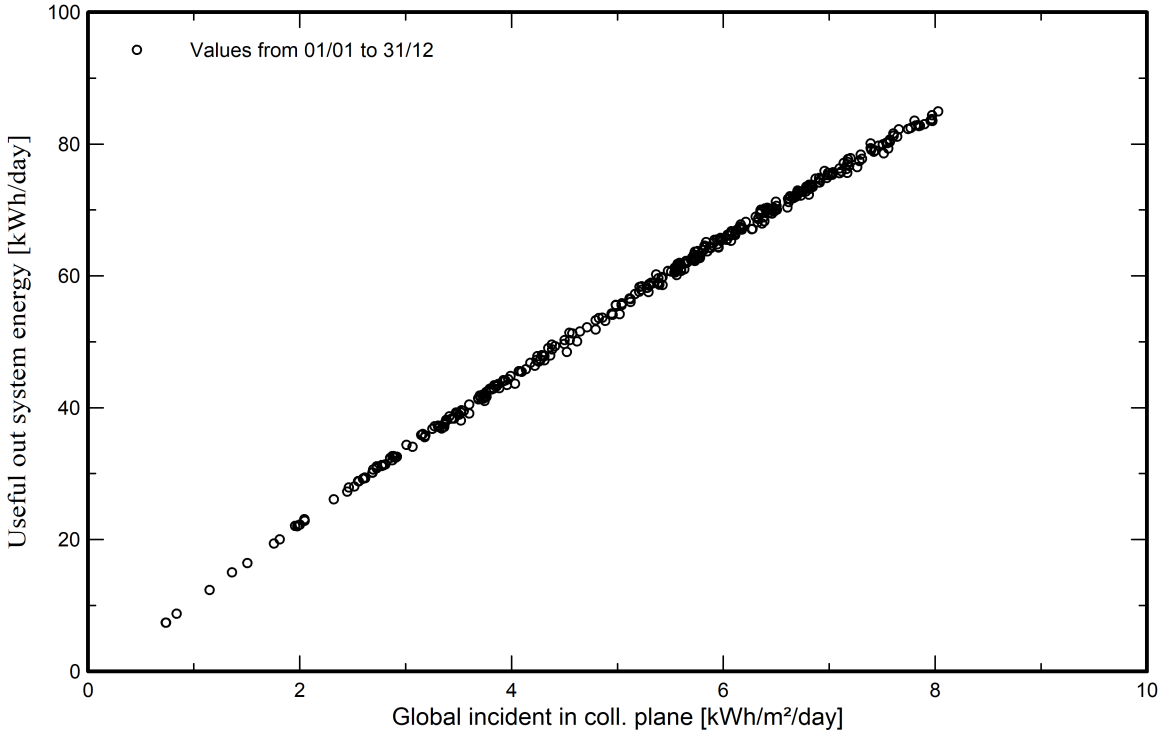
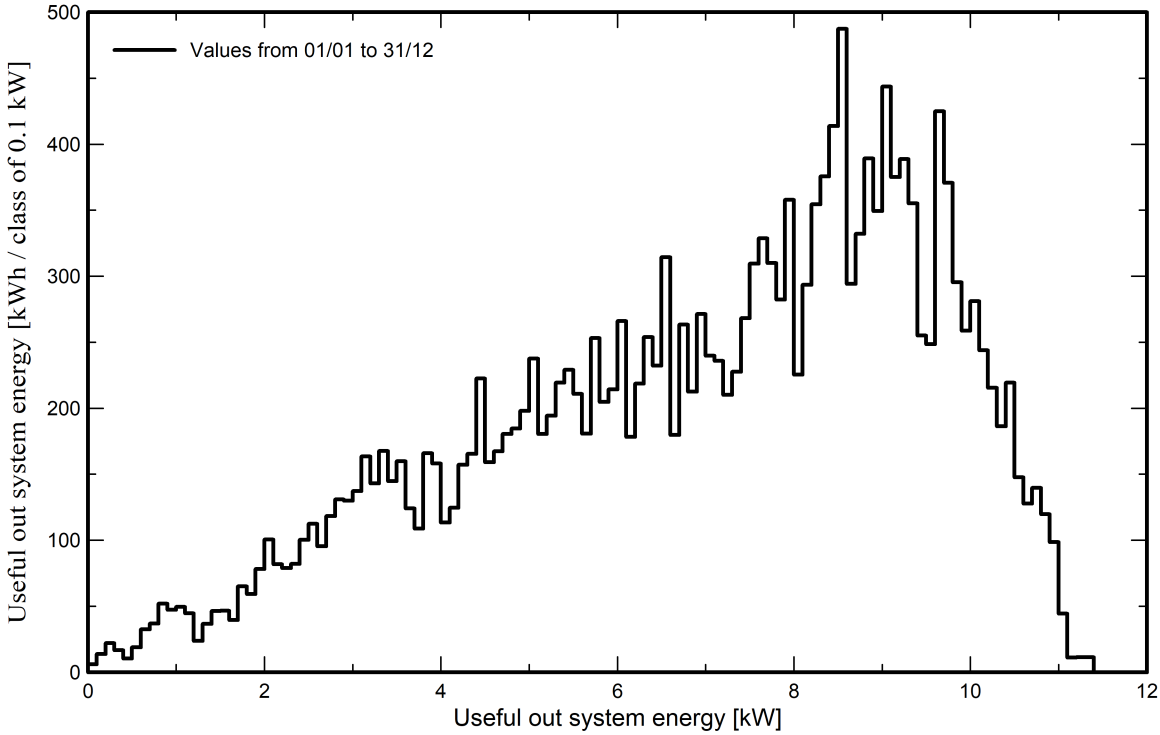
Predef. graphs

Daily Input/Output diagram



System Output Power Distribution



Single-line diagram

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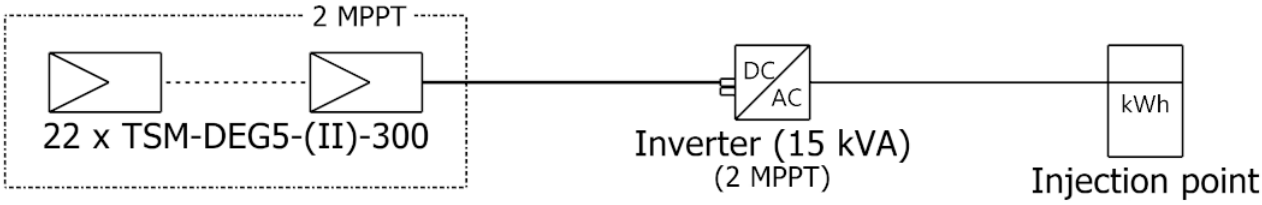

PV module	TSM-DEG5-(II)-300
Inverter	SUN2000-15KTL-M2
String	22 x TSM-DEG5-(II)-300

New Project

VC0 : New simulation variant

26/11/25